

User feedback: participant record

Purpose: improve assembly, operation and documentation. Participation is voluntary; stop or skip at any time. No names, student numbers or contact details are collected. Obtain separate permission for photos, recordings or public quotations and follow institutional requirements for participant studies.

Field	Response
Anonymous ID / date	_____
Manual language / file hash	ZH / EN;_____
Background	Biology / EE / mechanical / other_____
3D assembly experience	None /1-3 projects /4+
Prior AeroSense exposure	None / read / assembled
Board used	Dummy / real;_____
Participation	Understands right to stop:yes / no

Use dry parts and dummy/cell-free samples without pressurized gas. Use clearly labeled demo mode for UI tasks unless a separate verified hardware protocol applies. Observers avoid unsolicited coaching, respond to requested help and record it. Never force a part or take a risk.

Task log and ratings

Task	Seconds / completion / help / errors
Find all parts using BOM	_____
Fit fan, PCB and enclosure	_____
Identify wells and fit cap/tubes	_____
Open, replace a well, close	_____
Connect UI, start and stop	_____
Recognize DEMO/invalid; export	_____

Rate 1 strongly disagree, 2 disagree, 3 neutral, 4 agree, 5 strongly agree, or N/A. These are project-specific questions, not a validated SUS questionnaire; do not report a SUS score.

Item	1 2 3 4 5 N/A
I can identify each part and orientation	_____
Instructions/images are sufficient	_____
Fasteners/tools are accessible	_____
Replacing a well needs no excessive force	_____
Lid motion avoids tube pull/well contact	_____
I can identify inlet and exhaust	_____
UI state/errors are understandable	_____
I can distinguish demo/hardware data	_____
I would assemble again using this guide	_____

Open feedback and analysis plan

Least clear step/figure:_____

Hardest part / likely assembly error:_____

Which part/purchase could be removed, and why?_____

Which action raised collision/leak/contamination concerns?_____

One improvement to prioritize:_____

Other comments:_____

Observer logs damage, reversed parts, incomplete seating, missing washers, overtightening, tube strain and help requests. Predefine completion: all parts present/oriented, wells removable, connectors accessible and no visible mechanical load. Appearance alone is not completion.

Analysis: report participant backgrounds, completion rates, median/IQR time and error/help categories. Report per-item Likert distributions; N/A is not neutral. A five-novice pilot is formative, not proof of broad usability. Retest improvements with new participants to reduce learning effects.

For the wiki, trace feedback → design decision → retest result. Retain negative feedback and do not fabricate satisfaction, participant counts or questionnaire results. Anonymize quotations before publication.